



**SWAMINARAYAN
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THE RESERVOIR OF ALL KNOWLEDGE

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Chapter -1
Introduction to computer networks and Internet

1. Networks Introduction to computer and Internet

Computer Networks

A computer network is a group of interconnected devices that can communicate and share resources. These devices can include computers, servers, smartphones, and other hardware. Networks can be classified based on their size and scope:

- **Local Area Network (LAN):** Covers a small geographic area, like a home, office, or building.
- **Wide Area Network (WAN):** Spans a large geographic area, often a country or continent, and connects multiple LANs.
- **Metropolitan Area Network (MAN):** Covers a city or a large campus.
- **Personal Area Network (PAN):** Connects devices within a person's workspace, typically within a range of a few meters.

Internet

The **Internet** is the largest and most well-known computer network, connecting millions of private, public, academic, business, and government networks worldwide. It uses a standard set of protocols called **TCP/IP** (Transmission Control Protocol/Internet Protocol) to ensure different networks can communicate with each other.

Key Concepts

- **Nodes and Hosts:** Devices on a network.
- **Packets:** Small units of data transmitted over a network.
- **Protocols:** Rules that govern data communication. Common protocols include HTTP (for web browsing), FTP (for file transfer), and SMTP (for email).
- **IP Address:** A unique identifier for each device on a network.
- **Routers and Switches:** Devices that direct data traffic within and between networks.

Types of Networks

- **Social Networks:** Platforms like Facebook and Twitter that connect people.
- **Mobile Networks:** Networks that provide connectivity to mobile devices.
- **Computer Networks:** Networks that connect computers and other devices for data sharing and communication.

1.1 Understanding of network and Internet

Network

A **network** is a collection of computers, servers, and other devices connected to each other to share data and resources. Networks can vary in size and scope, and they are typically categorized into different types:

- **Local Area Network (LAN):** Covers a small area like a home, office, or building. Devices in a LAN are usually connected via Ethernet cables or Wi-Fi.
- **Wide Area Network (WAN):** Spans a large geographical area, such as a city, country, or even globally. The Internet itself is a type of WAN.
- **Metropolitan Area Network (MAN):** Covers a larger area than a LAN but smaller than a WAN, such as a city or a large campus.

Internet

The **Internet** is a vast global network of networks that connects millions of devices worldwide. It is a collection of public and private networks that are interconnected using standardized communication protocols. The primary purpose of the Internet is to provide a platform for communication, collaboration, and information sharing.

Key Differences

- **Scope:** A network can be as small as two connected devices, while the Internet is a global network of interconnected networks.
- **Purpose:** Networks are designed to facilitate communication and resource sharing within a specific area, whereas the Internet connects networks globally to enable broader communication and information exchange.
- **Technology:** Networks use various technologies like Ethernet, Wi-Fi, and fiber optics, while the Internet relies on standardized protocols like TCP/IP to ensure different networks can communicate with each other.

1.2 The network edge

In computer networks, the **network edge** is the point where a device or local network connects to the internet or another external network. This is a crucial area for both performance and security. Here are some key aspects:

1. **Connection Point:** The network edge is where devices like personal computers, smartphones, and IoT devices connect to the internet¹.
2. **Security Boundary:** It acts as a security boundary, protecting the internal network from external threats.

3. **Data Processing:** Edge computing often occurs here, processing data close to its source to improve response times and reduce bandwidth usage.
4. **Components:** Common components at the network edge include routers, switches, firewalls, and modems.

1.3 The network core

The **network core**, also known as the backbone network, is the central part of a computer network that connects various subnetworks and ensures efficient data transfer between them. Here are some key aspects:

1. **Connectivity and Routing:** The core network provides connectivity and routing services between different parts of the network, ensuring data packets are efficiently forwarded from one point to another.
2. **High Speed and Reliability:** It is designed to be very fast, reliable, scalable, and secure, handling large volumes of data with minimal latency.
3. **Components:** The core network typically includes high-capacity routers and switches that manage data traffic between different network segments.
4. **Packet and Circuit Switching:** It supports both packet switching, where data is divided into packets and sent independently, and circuit switching, where a dedicated path is established for the duration of a communication session.

1.4 Understanding of Delay

In computer networks, delay refers to the time it takes for data to travel from the source to the destination. There are several types of delays that can affect network performance:

1. **Transmission Delay:** This is the time required to push all the packet's bits onto the wire. It depends on the packet's length and the transmission rate of the link. For example, if the bandwidth is 1 Mbps and the packet size is 1,000 bits, the transmission delay is 1 millisecond.
2. **Propagation Delay:** This is the time it takes for a signal to travel from the sender to the receiver. It depends on the distance between the two and the propagation speed of the medium (e.g., fiber optics, copper wire). For instance, in fiber optics, the propagation speed is approximately (2×10^8) meters per second.
3. **Queueing Delay:** This occurs when packets wait in a queue before being transmitted. It varies based on the network traffic load. High traffic can lead to longer queueing delays.
4. **Processing Delay:** This is the time taken by routers and switches to process the packet header and determine the packet's next hop. It depends on the processing power of the network devices.

1.5 Loss and Throughput in the packet

In packet-switched networks, **loss** and **throughput** are crucial metrics that impact network performance.

Packet Loss

Packet loss occurs when data packets traveling across a network fail to reach their destination. This can happen due to network congestion, faulty hardware, or other

issues. Packet loss directly reduces throughput because some data is never received and can't be counted as part of the total data transferred. Additionally, many transport layer protocols interpret packet loss as a sign of congestion and will reduce their transmission rate to avoid further congestion.

Throughput

Throughput refers to the amount of data successfully transferred from one point to another in a given time period. It is often measured in bits per second (bps). Throughput can be affected by various factors, including packet loss, network congestion, and the efficiency of the network protocols in use.

Relationship Between Loss and Throughput

The relationship between packet loss and throughput is inversely proportional. As packet loss increases, throughput generally decreases because more data needs to be retransmitted, and protocols may slow down the transmission rate to manage congestion.

1.6 switching network

In computer networks, **switching** is the process of directing data packets from one device to another within the same network or across different networks. This is done using devices called **switches**. Here are the main types of switching techniques:

1. Circuit Switching

- **Description:** Establishes a dedicated communication path between two devices for the duration of the communication session.
- **Use Case:** Traditional telephone networks.
- **Advantages:** Provides a constant data rate and predictable performance.
- **Disadvantages:** Inefficient for data transmission as the dedicated path can lead to wasted bandwidth.

2. Packet Switching

- **Description:** Data is broken into packets, each of which can take different paths to the destination. The packets are reassembled at the destination.
- **Use Case:** Most modern computer networks, including the internet.
- **Advantages:** Efficient use of network resources, supports multiple communication sessions simultaneously.
- **Disadvantages:** Potential for packet loss and delay.

3. Message Switching

- **Description:** Entire messages are sent from the source to the destination, one hop at a time. Each intermediate device stores the entire message before forwarding it.
- **Use Case:** Early data networks.
- **Advantages:** No need for a dedicated path, can handle large messages.

- **Disadvantages:** High latency due to the store-and-forward nature.

Network Switches

A **network switch** is a hardware device that connects multiple devices within a network and uses MAC addresses to forward data to the correct destination. Switches operate at the Data Link layer (Layer 2) of the OSI model

1.7 protocols layers and their service model

In computer networking, protocols are organized into layers, each with specific functions and services. The most well-known models are the **OSI (Open Systems Interconnection) model** and the **TCP/IP model**. Here's a brief overview of each layer and its service model:

OSI Model

The OSI model has seven layers, each providing specific services to the layer above it:

1. **Physical Layer**
 - **Function:** Transmits raw bit streams over a physical medium.
 - **Protocols:** Ethernet, USB.
2. **Data Link Layer**
 - **Function:** Provides node-to-node data transfer and error detection/correction.
 - **Protocols:** MAC (Media Access Control), PPP (Point-to-Point Protocol).
3. **Network Layer**
 - **Function:** Handles routing of data packets between devices across different networks.
 - **Protocols:** IP (Internet Protocol), ICMP (Internet Control Message Protocol).
4. **Transport Layer**
 - **Function:** Ensures complete data transfer with error recovery and flow control.
 - **Protocols:** TCP (Transmission Control Protocol), UDP (User Datagram Protocol).

5. **Session Layer**
 - **Function:** Manages sessions or connections between applications.
 - **Protocols:** NetBIOS, RPC (Remote Procedure Call).
6. **Presentation Layer**
 - **Function:** Translates data between the application layer and the network, including encryption and compression.
 - **Protocols:** SSL/TLS (Secure Sockets Layer/Transport Layer Security).
7. **Application Layer**
 - **Function:** Provides network services directly to end-user applications.
 - **Protocols:** HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), SMTP (Simple Mail Transfer Protocol).

TCP/IP Model

The TCP/IP model, which is more commonly used in practice, has four layers:

1. **Link Layer**
 - **Function:** Corresponds to the OSI's Physical and Data Link layers, handling hardware addressing and media access control.
 - **Protocols:** Ethernet, ARP (Address Resolution Protocol).
2. **Internet Layer**

- **Function:** Corresponds to the OSI's Network layer, responsible for packet forwarding and routing.
- **Protocols:** IP, ICMP.
- 3. **Transport Layer**
 - **Function:** Similar to the OSI's Transport layer, ensuring reliable data transfer.
 - **Protocols:** TCP, UDP.
- 4. **Application Layer**
 - **Function:** Combines the OSI's Session, Presentation, and Application layers, providing end-user services.
 - **Protocols:** HTTP, FTP, SMTP.

Service Models

Each layer provides services to the layer above it. For example:

- The **Network Layer** provides routing services to the **Transport Layer**.
- The **Transport Layer** provides reliable data transfer services to the **Application Layer**.

These models help standardize network communication and make it easier to develop and troubleshoot networked systems

1.8 History of the computer network

The history of computer networking is fascinating and spans several decades, marked by significant milestones and technological advancements. Here's a brief overview:

Early Developments

- **1961:** Leonard Kleinrock proposed the idea of ARPANET, one of the earliest computer networks, in his paper "Information Flow in Large Communication Nets"¹.
- **1965:** Donald Davies coined the term "packet" to describe data sent between computers over a network¹.
- **1969:** ARPANET, the first network to use packet switching, was developed. The first data transmission occurred between UCLA and the Stanford Research Institute¹.

Key Milestones

- **1973:** Robert Metcalfe developed Ethernet at Xerox PARC, which became a foundational technology for local area networks (LANs)¹.
- **1978:** Bob Kahn and Vint Cerf developed the TCP/IP protocol, which became the standard for internet communication¹.
- **1983:** ARPANET transitioned to using TCP/IP, marking the beginning of the modern internet¹.
- **1986:** The National Science Foundation Network (NSFNET) was established, providing a backbone for ARPANET and eventually replacing it.

Modern Era

- **1990s:** The World Wide Web was invented by Tim Berners-Lee, revolutionizing how information is accessed and shared².
- **2000s:** The rise of wireless networking technologies, such as Wi-Fi, further expanded the reach and convenience of computer networks

